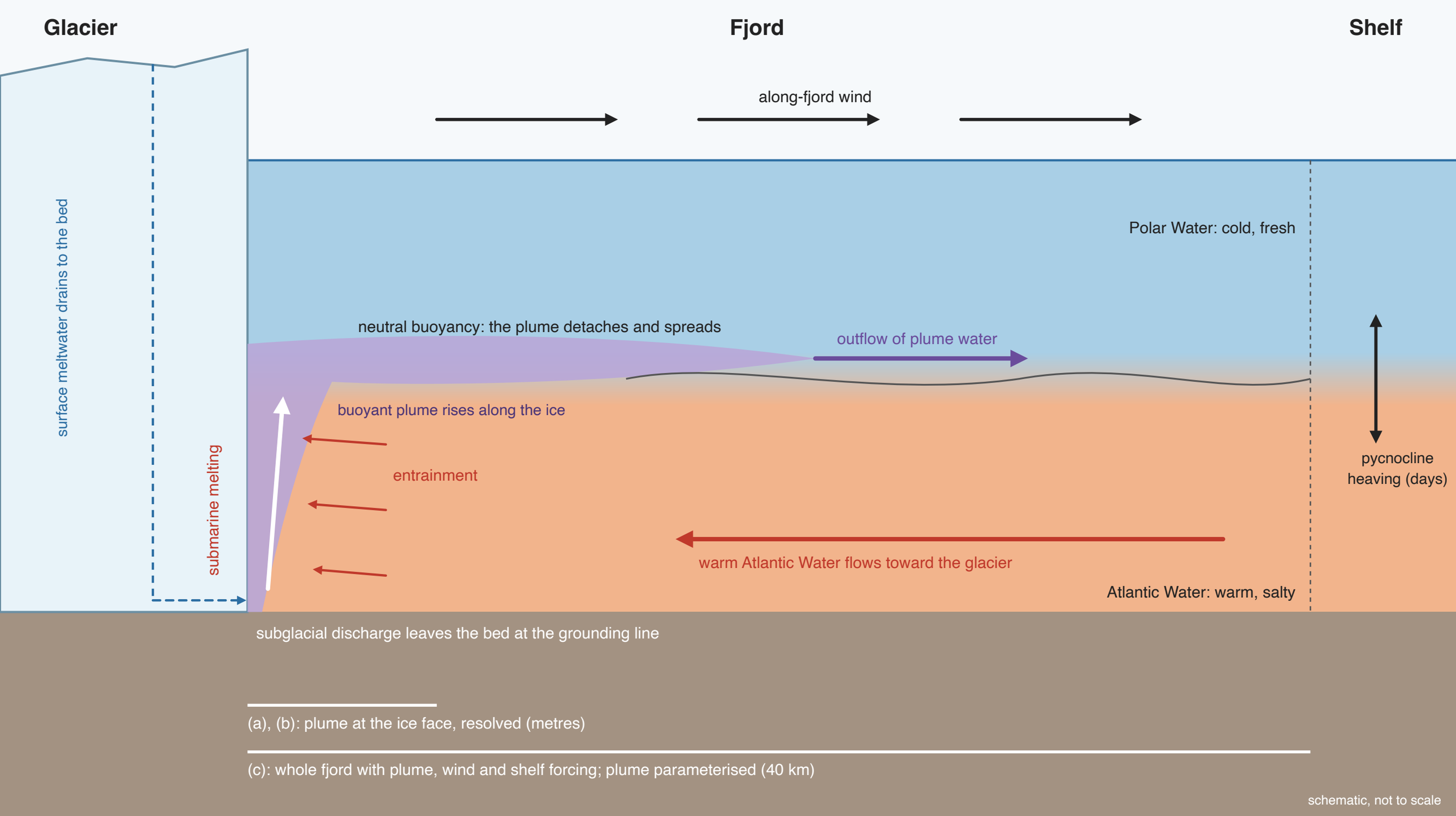
Glacier
Fjord
Shelf
along-fjord wind
surface meltwater drains to the bed
submarine melting
Polar Water: cold, fresh
neutral buoyancy: the plume detaches and spreads
outflow of plume water
buoyant plume rises along the ice
entrainment
pycnocline
heaving (days)
warm Atlantic Water flows toward the glacier
Atlantic Water: warm, salty
subglacial discharge leaves the bed at the grounding line
(a), (b): plume at the ice face, resolved (metres)
(c): whole fjord with plume, wind and shelf forcing; plume parameterised (40 km)
schematic, not to scale